Machine Learning Laboratory Assignment

Overfitting & Underfitting | Bias–Variance Tradeoff | Logistic Regression

Program: M.Tech – 1st Year Course: Machine Learning Type: Practical / Lab Assignment

Total Questions: 4 (Section A: 1, Section B: 1, Section C: 2)

General Instructions

- Use Python with NumPy, Pandas, Matplotlib, and scikit-learn. For "from scratch" parts, use only NumPy/Pandas for the core algorithm; scikit-learn may be used afterwards for comparison where the question asks for it.
- Fix the random seed (random_state = 42) wherever randomness is involved, for reproducibility.
- Every figure must have a title, labelled axes, and a legend where more than one curve is plotted.
- Select all hyperparameters using a validation split or cross-validation (for regression use k-fold CV, for classification use stratified k-fold CV) — never by tuning against the test set.
- Add a short "Observations / Inference" note (2-3 sentences) after each question interpreting your results.
- Question 1, 2, 3 are compulsory and question 4 is optional.
- Copied code or shared solutions between students will be treated as plagiarism.

Section A: Overfitting and Underfitting (Question 1)

Question 1: Visualizing Overfitting and Underfitting via Polynomial Curve Fitting

Dataset / Setup: Synthetic data from $y = \sin(2\pi x) + \varepsilon$, $\varepsilon \sim N(0, 0.2^2)$: 25 training points and a separate, larger (e.g., 300-point) test set, both with $x \sim \text{Uniform}(0, 1)$.

Tasks

(a) Generate the training and test sets as described above. Fit unregularized polynomial regression models for degree $d \in \{1, 4, 15\}$, and plot each fitted curve along with the training points and the true function $\sin(2\pi x)$ on the same axes.

(b) Compute the training and test MSE for each of the three degrees and tabulate the results.

(c) Based on your results, identify which degree underfits, which gives a good fit, and which overfits, with a brief justification for each.

Section B: Bias–Variance Tradeoff (Question 2)

Question 2: Empirical Bias^2–Variance Decomposition via Polynomial Degree

Dataset / Setup: Same data-generating process as Question 1: $f(x) = \sin(2\pi x)$, $y = f(x) + \varepsilon$, with $\sigma = 0.2$ (irreducible error $\sigma^2 = 0.04$). Fix a 100-point evaluation grid in [0, 1], and generate $B = 100$ independent training sets ($N = 30$ each) by drawing fresh x and ε every time.

Tasks

(a) For each degree $d \in \{1, 4, 15\}$, fit a degree-d polynomial model on every one of the 100 training sets and predict on the fixed evaluation grid.

(b) At each grid point, compute the average prediction, the squared bias (against the true f(x)), and the variance across the 100 models; average these over the grid to get overall $\text{Bias}^2(d)$ and Variance(d) for each degree.

(c) Plot Bias^2, Variance, and their sum (Bias^2 + Variance) versus degree on one graph, and identify which degree gives the best balance between the two.

Section C: Logistic Regression (Questions 3–4)

Question 3: Logistic Regression from Scratch using Batch Gradient Descent

Dataset / Setup: A 2-feature synthetic dataset (sklearn.datasets.make_classification, n_samples=300, n_features=2, n_redundant=0, random_state=42), split 80:20.

Tasks

(a) State the sigmoid function $\sigma(z)$ and the cross-entropy cost function $J(w)$ used in logistic regression.

(b) Implement the sigmoid, the cost function, and batch gradient descent ($w \leftarrow w - \eta \nabla J(w)$) in NumPy; train the model on the dataset and plot the cost versus iteration.

(c) Plot the training points along with the learned decision boundary, and report the training accuracy.

Question 4: Regularized Logistic Regression: L1 versus L2 **(OPTIONAL)**

Dataset / Setup: Synthetic dataset via make_classification(n_samples=400, n_features=20, n_informative=5, n_redundant=5, random_state=42), standardized, split 70:30.

Tasks

(a) Fit L2-regularized and L1-regularized logistic regression for $C \in \{0.01, 1, 100\}$; plot test accuracy versus C for both penalty types.

(b) For the L1 models, also plot the number of non-zero coefficients versus C, and briefly comment on how L1 performs feature selection as C increases.

(c) Using cross-validation, select the best C for each penalty type and report the corresponding test accuracy, precision, and recall.